

Structure and Function of the Respiratory System & Respiratory Disorders

Integrated Biomedical and Clinical Sciences OMFS 512

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Learning Objectives

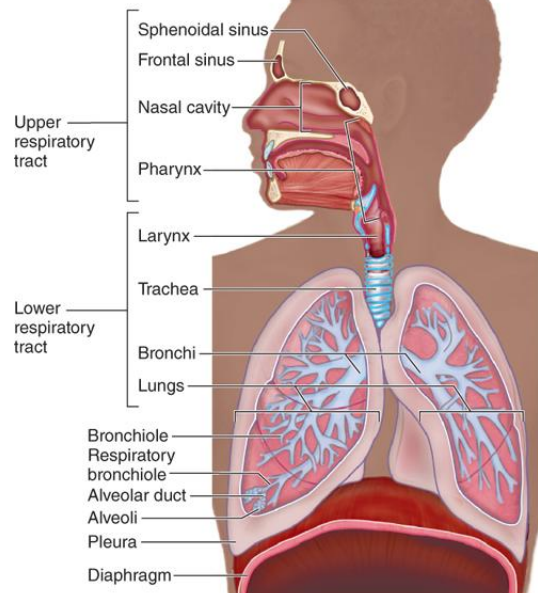
Learning Outcomes

1. Describe pulmonary system, pulmonary circulation and neural innervation.
2. Discuss the physiological function of the respiratory system.
3. Breathing cycle and control of breathing.
4. Describe disorders associated with pulmonary system.

Outline

- Development of the respiratory system
- Respiratory circulation and neural innervation
- Breathing cycle and control of breathing
- Developmental disorders.
- Ventilation disorders.
- Vascular disorders.
- Pulmonary Diseases: Asthma.

Development of Respiratory System



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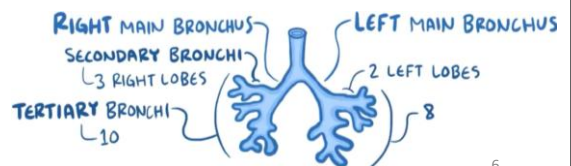
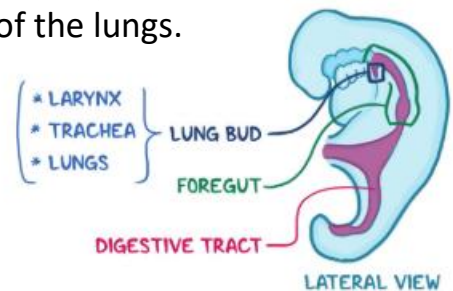
Embryology

Endoderm develop into the mucosal lining of bronchi & epithelial cells of alveoli.
Mesoderm give rise to the remaining components of the lungs.

Five stages:

1- Embryonic stage (week 4) The lung bud originates from the foregut to yield lower respiratory tract structures.

2- Pseudoglandular stage (weeks 5-16)
 The conducting airways progressively branch forming bronchi and terminal bronchioles.

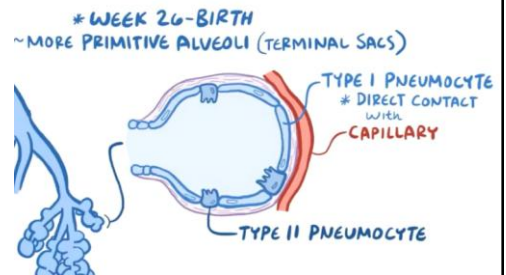
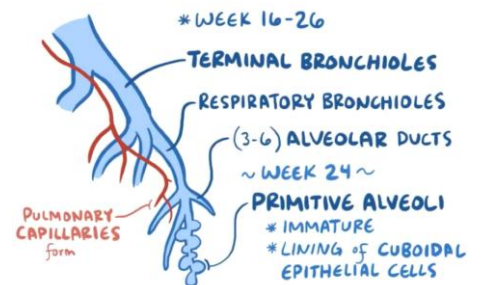


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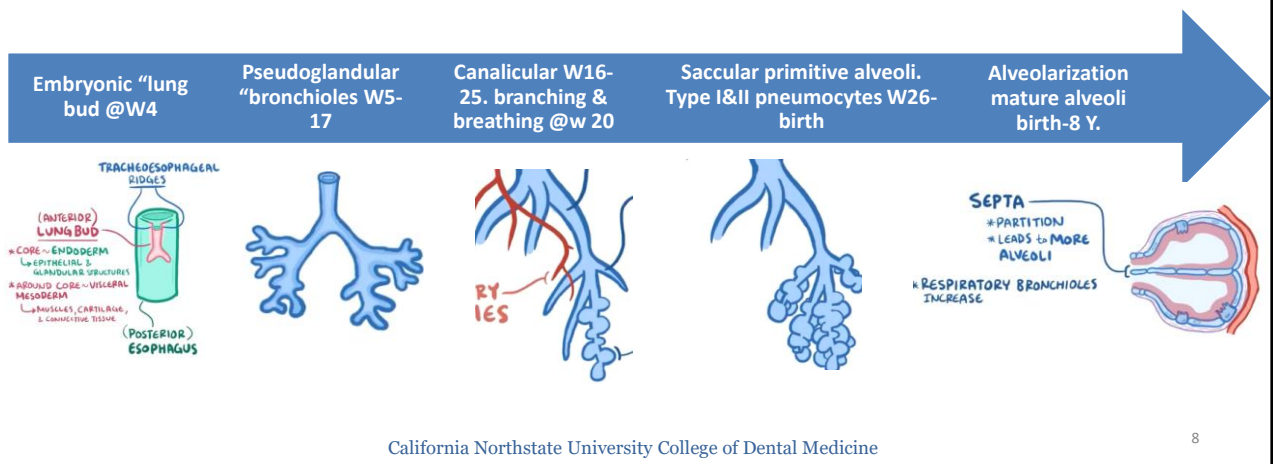
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3- Canalicular stage (week 16-25): continuous branching & formation of respiratory bronchioles. Week 20: fetal breathing movements may begin & epithelium differentiates into type I pneumocytes, responsible for gas exchange.

4- Saccular stage (week 26 to birth). Maturation of type I and type II pneumocytes and formation of primitive alveoli (sacculi).

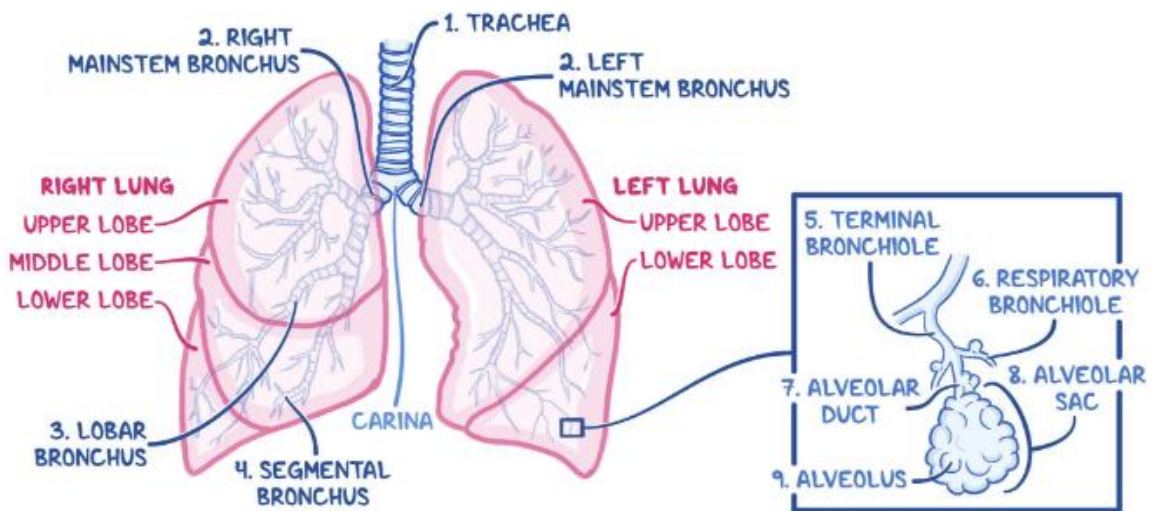


5- Alveolarization stage (birth till 8 yrs). Respiratory bronchioles and alveoli increase in number. Around 15 % of total number of adult alveoli are present at birth (70 mil. at birth to 400 million at 8 yrs). Septation: septa partition the primitive alveoli to yield more adult alveoli with larger surface area for efficient gas exchange.



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Foreign body aspiration. More common in children <3 Yrs. Adults >60 yrs (dental material/prosthesis 40 % cases). Foreign particles can be aspirated into the right lung (right bronchus; 65 % cases) in > left lung (bronchus; 30 % cases). Symptom range from asymptomatic to cough, dyspnea, wheezing.....asphyxia & death.

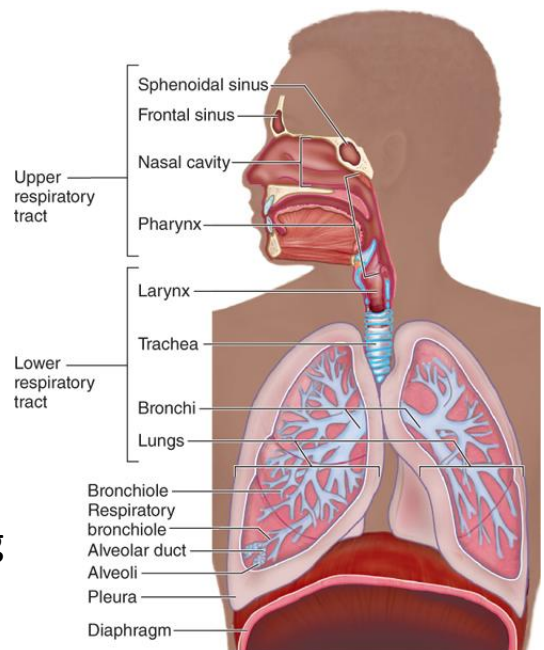
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Respiratory System

Anatomically: divided into the upper and lower respiratory tracts within thoracic cavity. Lungs enclosed in pleura.

Functionally: divided into a conducting portion and a respiratory portion



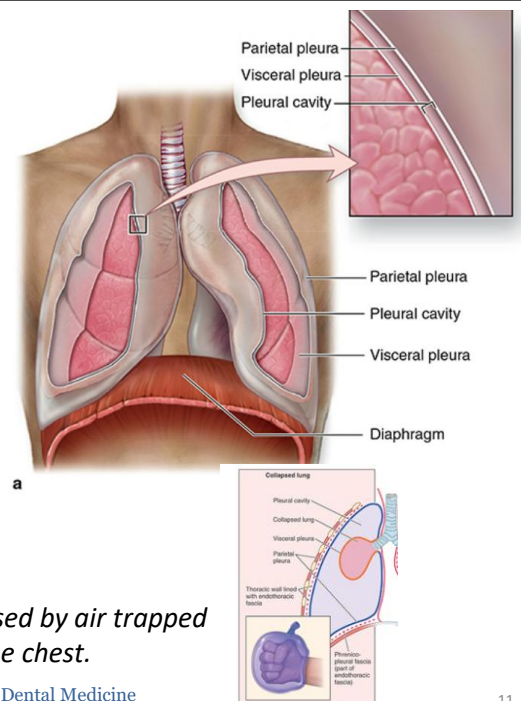
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Pleura

Lungs are covered by a thin double-layered serous membrane “pleura”.

- a) Parietal pleura: outer pleural layer, adherent to the thoracic wall.
- b) Visceral pleura: inner pleural layer, adjacent to the lung.
- c) Pleural cavity: space in-between, filled by pleural fluid that acts as a lubricant. Pleural effusion= fluid buildup in the pleural cavity. shortness of breath.



Pneumothorax is a partially or completely collapsed lung caused by air trapped in the pleural cavity. Can result from penetrating trauma to the chest.

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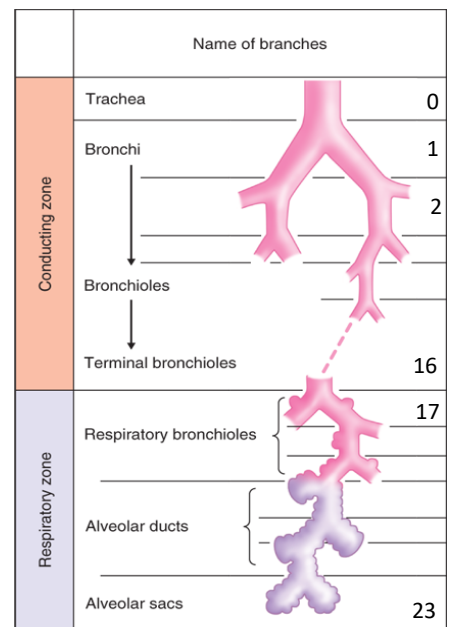
Zones of the airway

Conducting zone:

Structures/airways that deliver air to the lungs.

Respiratory zone:

Oxygen/carbon dioxide exchange.
Generations 17-23.



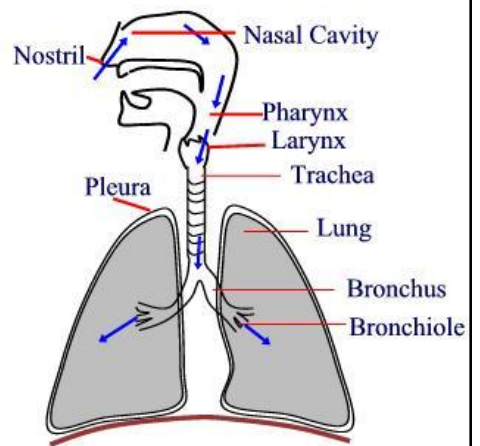
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Source: K.E. Barrett, S.M. Barman, H.L. Brooks, Jason X.J. Yuan:
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Conducting zone

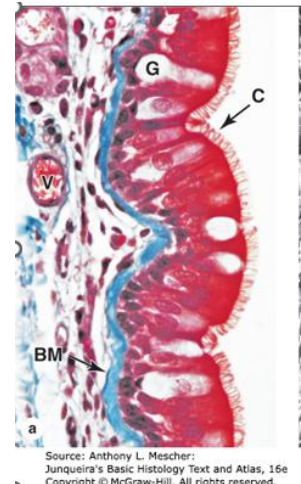
- From nose, nasal sinus, to terminal bronchi.
- The first 16 generation of bronchial tree (bronchi, bronchioles, and terminal bronchioles).
- Functions for air conditioning (moisten & warm), filtration and olfaction (odors & inhaled irritant).



[Biology : Human Respiratory system - Worksheet / Test Paper \(syvum.com\)](#)

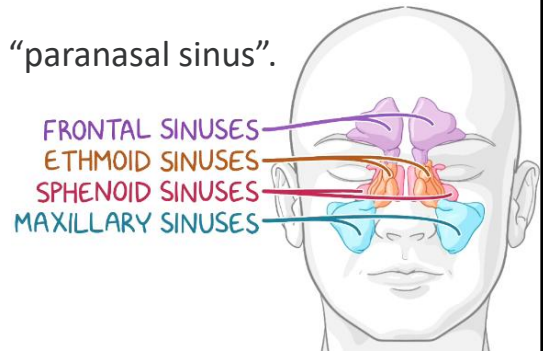
Distinct cell population in conducting portion:

- **Pseudostratified ciliated columnar cells** most abundant population (c).
- **Goblet cells** (G) numerous in number, responsible for the secretion of mucus. Mucus contains antimicrobial enzymes (like lysozymes), salts, water and mucin glycoprotein (lubricant). Hyperplasia of goblet cell can be associated with asthma and COPD.
- **Basal cells** (BM) are mitotically active progenitor cells that give rise to the other epithelial cell types.



Paranasal sinuses

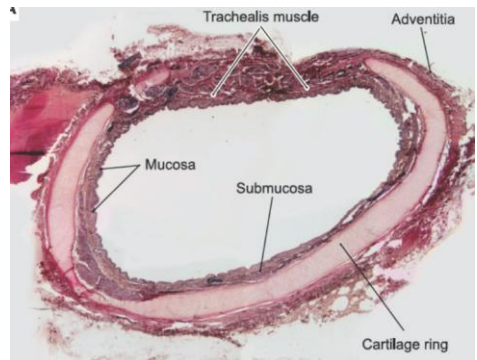
- The nasal cavity is connected to four sinuses “paranasal sinus”.
- Four paired air-filled spaces surrounding nasal cavity.
- Lighten the weight of the head; humidifying inspired air; increase resonance of the voice, supporting immune defense of the nasal cavity; produces mucus that drains into the nasal cavity.



Trachea

The trachea is a fibromuscular tube supported by C-shaped hyaline cartilage in the anterior portion and a trachealis smooth muscle at the dorsal part (unique to trachea).

Lined with respiratory epithelia. Ciliated cells that beat rhythmically to move mucus and any trapped particles towards the pharynx. This clearing mechanism is referred to as “mucociliary escalator”.

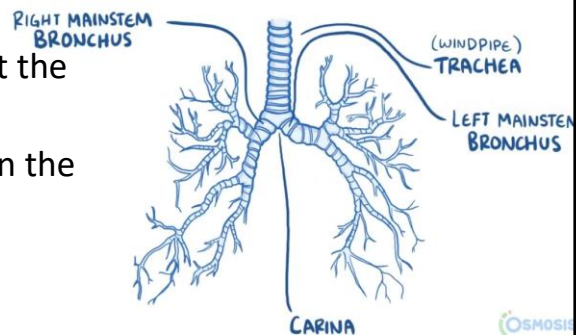


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Bronchi and bronchioles

- The trachea divides into two primary bronchi at the carina.
- The right and left bronchi enter each lung at the hilum.
- The right bronchus is shorter and wider than the left bronchus.
- Walls of the bronchi and bronchioles are innervated by autonomic nervous system affecting width and mucus secretion.



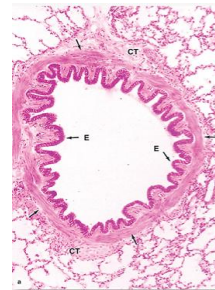
Bronchi vs bronchioles

Bronchi: smooth muscle layer (SM), cartilage (C) and submucosal glands (G). Goblet cells to secrete mucus.



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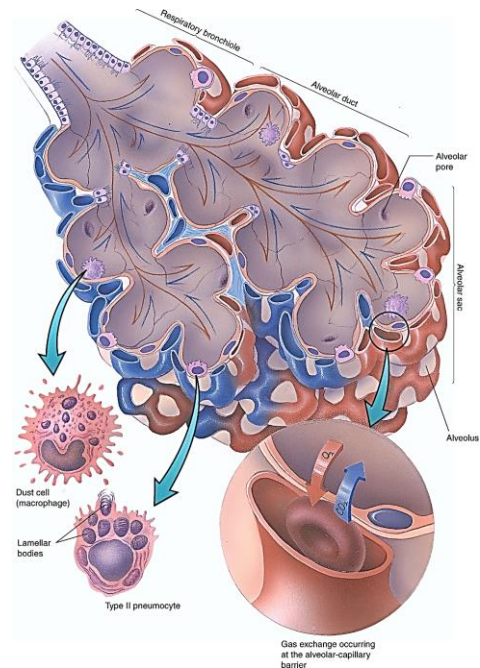
Bronchioles : smooth muscle layer but no cartilage and no submucosal glands. Club (aka, clara) cells: to secrete mucin, lipoprotein surfactant (to reduce surface tension) and antimicrobial peptides.



Respiratory zone

For gas exchange. It include the respiratory bronchioles, alveolar ducts, alveolar sacs, and alveoli.

- Alveoli are pouch-like structures located in the walls of alveolar sacs. Functional units of the respiratory system.
- Alveoli have thin walls, across which oxygen and carbon dioxide exchange between air and the blood



Components of the respiratory portion of the respiratory system
BRS Cell Biology and Histology, (From Gartner LP. *Color Atlas of Histology*. 7th ed. Baltimore, MD: Wolters Kluwer; 2018:331.)

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Alveolar cells

Alveoli are lined by simple squamous epithelium composed of equal number of types I and II pneumocytes or alveolar cells.

1- **Type I pneumocytes** flat epithelial cells.

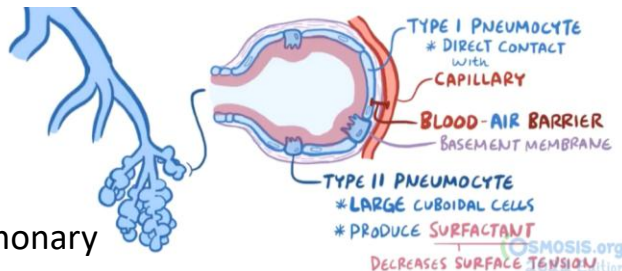
They are critical for the gas exchange.

Occupy 95 % of the alveoli surface area.

Doesn't divide.

2- **Type II pneumocytes**

cuboidal epithelial cells. Produce the pulmonary surfactant & serve as stem cells to replace damaged type I pneumocytes.



3- Alveolar macrophages (aka, dust cells or heart failure cells)

Monocytes migrate function to phagocytose excess surfactant or erythrocytes from damaged capillaries, & airborne particulate matter that penetrated the alveoli.

Interalveolar septa: between neighboring alveoli.

Consists of scattered fibroblasts and extracellular matrix (ECM; elastic and reticular fibers, of connective tissue). Elastin in elastic fibers enables elastic recoil in alveoli (expand & contract with inspiration & expiration, respectively). Damage/loss associated with COPD as emphysema

Lung surfactant

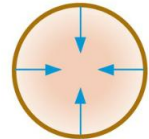
Responsible for maintaining the lung elasticity.

- Reduces surface tension.
- Increases pulmonary compliance (easier expansion/stretching).
- Prevents the collapse of alveoli during expiration.

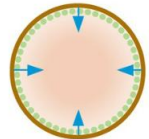
Composed of

- lipoproteins complex (dipalmitoyl phosphatidylcholine or DPPC),
- cholesterol
- four surfactant proteins (SP-A to D).
- Production starts at W26 gestation. Facilitates transition from fluid-filled lung in utero to an air-filled lung shortly after delivery.

Small alveolus



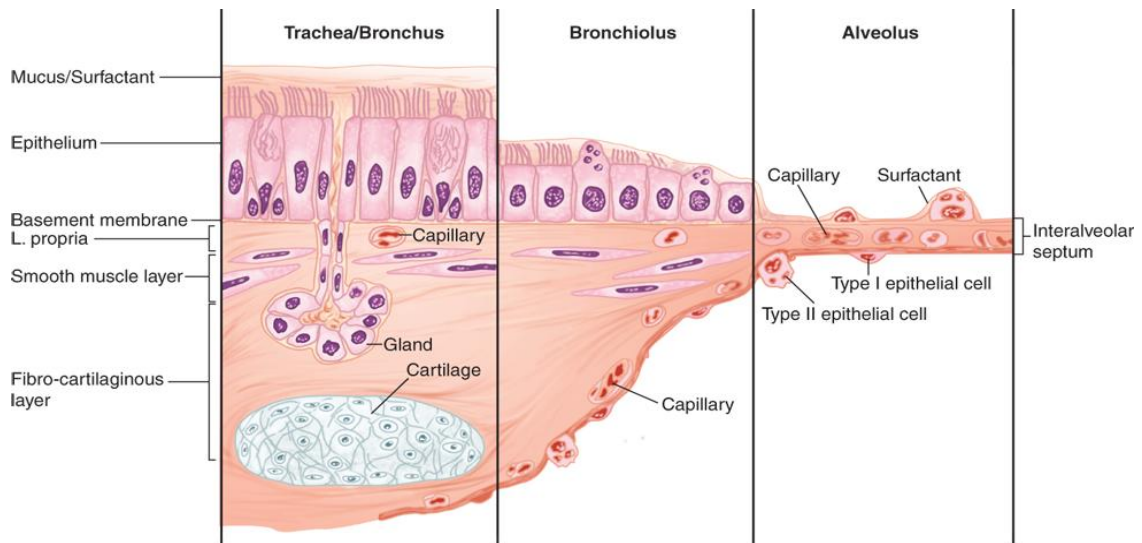
Small alveolus with surfactant



- Lack of pulmonary surfactant is responsible for **respiratory distress syndrome** in premature infants.
 - **Neonatal respiratory distress syndrome (RDS)**, in premature infants. Symptoms include labored breathing. Can be prevented by administration of glucocorticoids to mom to induce the synthesis of surfactant. Alternatively, synthetic DPPC is administered to neonates (<24 weeks of gestation).
- Defects on the lung surfactant play an important role in obstructive lung diseases.
- Type II pneumocytes are targets for SARS-CoV-2. Infected cells have reduced surfactant production, leading to alveolar collapse, and variable levels of hypoxia

Adaptation

Organization, cell structure, function



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Blood supply and Neural Innervation

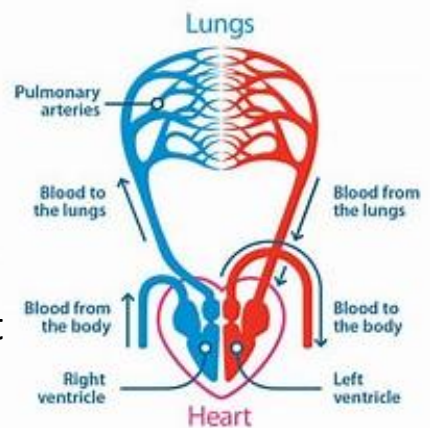
Functional and nutritive circulation

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Pulmonary Circulation

- Pulmonary circulation: functional circulation of the lung that aims to reoxygenate the blood and release carbon dioxide. Receives 100% of the cardiac output.
- Deoxygenated blood is pumped into the right and left pulmonary arteries to the right and left lungs, respectively. Oxygenated blood returns to the left atrium via the pulmonary veins.

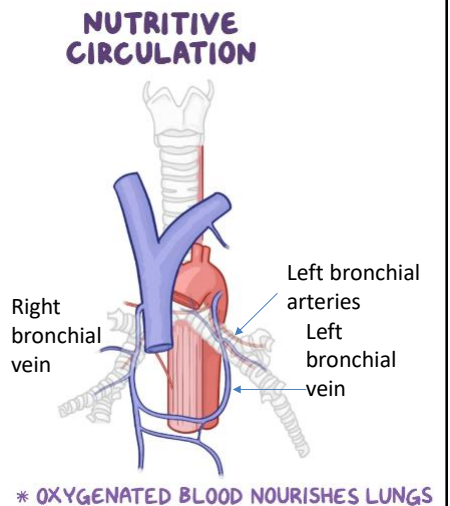


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Bronchial circulation

- Distinct circulation. Receives approximately 1% of total cardiac output.
- Bronchial artery: Supplies the lungs with systemic oxygenated blood.
- Bronchial vein: carry deoxygenated blood to the heart.



Autonomic innervation

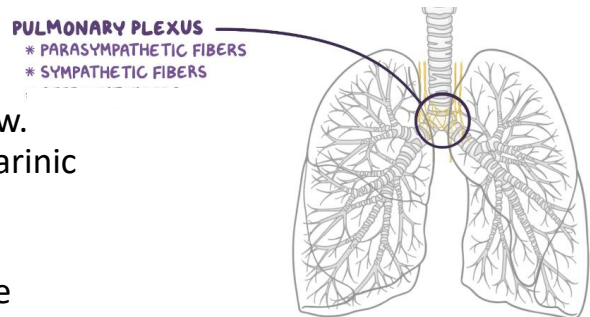
Parasympathetic stimulation

Stimulation constrict pulmonary smooth muscle & increase the resistance to airflow. Increase mucus secretion via Ach & muscarinic receptors.

At rest, vagus input is responsible for the slightly constricted normal smooth muscle tone.

Sympathetic stimulation

Stimulation dilate the airway smooth muscle (via β_2 receptors) and decrease the resistance to airflow. Constriction of blood vessels, and inhibition of glandular secretion.



Functions of the respiratory system

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- **Respiratory functions:**

- Oxygen and carbon dioxide exchange.
- Air conditioning.

- **Non respiratory functions:**

- Phonation
- Metabolic: sustaining aerobic metabolism & lung endothelia are main site for ACE production (converts angiotensin I to II).
- Maintain the acid-base balance through removal of carbon dioxide
- Immune function: Inhaled air is filtered by the conducting zone. The lung contain alveolar macrophages that phagocytose foreign particulates.

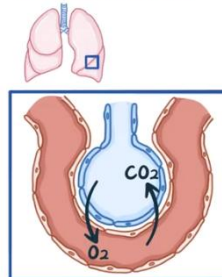
External and Internal respiration

Oxygen and carbon dioxide diffuse from the high pressure to low pressure.

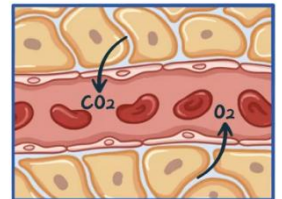
External respiration: Oxygen diffuse from alveoli (high P_{O_2}) into blood (low P_{O_2}) and carbon dioxide diffuse from blood (high P_{CO_2}) into alveoli.

Internal respiration Oxygen diffuse from the blood into tissues.

EXTERNAL RESPIRATION
- BULK FLOW of AIR INTO
& OUT of the LUNGS

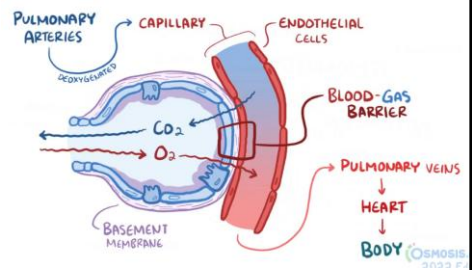


INTERNAL RESPIRATION
- CAPILLARY GAS EXCHANGE
in BODY TISSUES



Oxygen transport

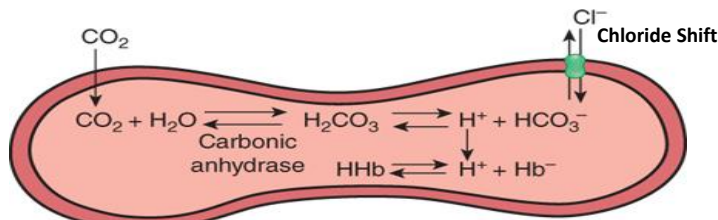
- O₂ is carried in blood in two forms: dissolved or bound to hemoglobin (most prominent).
- Gas exchange occurs between the alveoli & RBCs (of pulmonary capillaries) across the blood-gas interface (blood-gas barrier) via passive diffusion.
- The **blood-gas interface** is made up of type I alveolar cells, capillary endothelium and basement membrane that holds capillaries & alveoli together.



Carbon dioxide transport

In the RBCs, 70 % of carbon is converted to carbonic acid (H_2CO_3) by carbonic anhydrase.

- Carbonic acid dissociates into a bicarbonate anion (HCO_3^-) and hydrogen ion.
- Bicarbonate will diffuse to the plasma. *Plasma bicarbonate is the major carbon dioxide transport form.*
- Chloride Shift: To maintain ionic balance between RBCs and plasma, bicarbonate ion is exchanged with one chloride ion.
- In the lungs, process is reversed, and carbon dioxide is exhaled.



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Factors affecting gas exchange

- **Fick's law**: the net rate of diffusion of any gas across the blood-gas interface is:
 - directly proportional the driving force (difference between partial pressure of the gas in the alveoli and P_a of the gas in the blood)
 - directly proportional to surface area of the respiratory membrane (e.g: SA for gas exchange is decreased in pulmonary diseases that destroys alveoli like emphysema)
 - inversely proportional to the membrane thickness (normal thickness is 0.5-1 μm). Membrane thickness increases in pathologies like lung fibrosis.

Haldane Effect

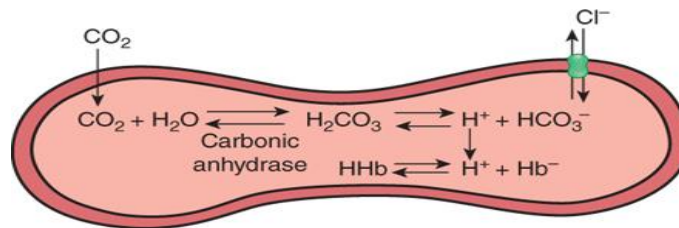
In the lungs, the binding of oxygen to Hb decreases its affinity for carbon dioxide. This aids in the release of carbon dioxide from Hb (carbaminohemoglobin) and facilitate subsequent oxygen binding (i.e, facilitates unloading of carbon dioxide and loading of oxygen in the lung).

Bohr's effect

In peripheral tissues. Describes hemoglobin-oxygen dissociation. Accumulation of H^+ , carbon dioxide, and 2,3-bisphosphoglycerate decreases affinity of Hb to oxygen and facilitates its release into tissues to meet their metabolic demand.

Regulation of acid-base balance

- Carbon dioxide elimination from the lungs can directly affect the acid base balance and plasma pH.
- Respiratory rate can affect the carbon dioxide level as well as bicarbonate (HCO_3^-) and hydrogen (H^+) ion levels.



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Respiratory Acidosis and Alkalosis

The pH of plasma is maintained within a narrow range of 7.38 to 7.42.

Hypoventilation leads to a decrease of exhaled carbon dioxide, eventually leading to increased H^+ levels and acute respiratory acidosis. Acidosis defines a plasma pH below 7.3. Acidosis is life threatening if the pH goes below 7.25.

Alternatively, acute or chronic conditions that lead to an increase of carbon dioxide exhalation will lower the H^+ concentrations leading to respiratory alkalosis. Alkalosis defines a plasma pH more than 7.4. Alkalosis is symptomatic at pH above 7.55.

Renal compensation in acid-base disorders

A) Acidosis:

- 1- Metabolic acidosis due to increased acid production (like ketoacidosis associated with diabetes) and renal failure.
- 2- Respiratory acidosis due to hypoventilation (opiates or sedative overdose, COPD).

Renal compensation consists of increased excretion of H^+ and increased reabsorption of HCO_3^- .

B) Alkalosis:

1- Metabolic alkalosis due to vomiting, diuretics toxicity.

Renal compensation consists of increased excretion of HCO_3^- .

2- Respiratory alkalosis due to hyperventilation.

Renal compensation consists of decreased excretion of H^+ and decreased reabsorption of HCO_3^- (decrease serum bicarbonate level).

In respiratory acidosis and alkalosis, renal compensation occur in chronic conditions.

Breathing physiology

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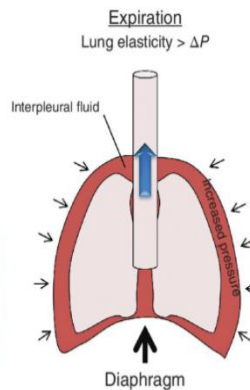
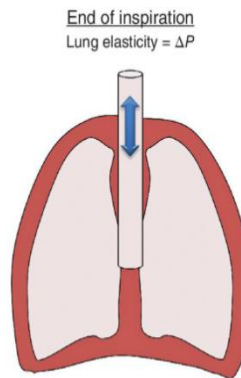
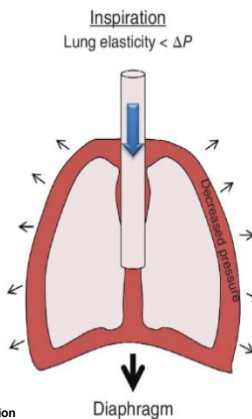
Pulmonary ventilation refers to the airflow into and out of the lungs. Normal breathing rate in adults is between 12 to 18 breaths per min.

Term	Definition
Eupnea	Normal, unlabored breathing (when the body is at rest).
Dyspnea	Difficulty breathing or shortness of breath
Orthopnea	Shortness of breath in supine position. Relieved by standing or sitting
Apnea	Cessation of breathing
Hyperpnea	increased depth and rate of breathing
Tachypnea	Refers to rapid rate of shallow breathing>20 breaths/min.

Breathing Cycle

Ventilation is facilitated by the pressure difference between the atmosphere and alveoli to enable the movement of air into or out of the alveoli. Intrapleural pressure is negative

Contraction: **diaphragm's downward movement** and external intercostal muscles contraction create negative pressure and expand the thoracic cage.



Relaxation: **diaphragm** and the internal intercostal muscle relax. Lung surfactant facilitate the elastic rebound of the alveoli to regain their original shape without collapsing.

Mechanics of expiration and inspiration
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A. Muscles of inspiration:

1. Diaphragm: most important muscle for inspiration.
2. External intercostals and accessory muscles are mainly used for inspiration during exercise and in respiratory distress.

B. Muscles of expiration

Generally passive in nature. Expiratory muscles are used during exercise or pulmonary diseases that increase resistance (like, asthma).

1. Abdominal muscles to compress the abdominal cavity, push the diaphragm up to push air out of the lungs.
2. Internal intercostal muscles that pull the ribs downward and inward.

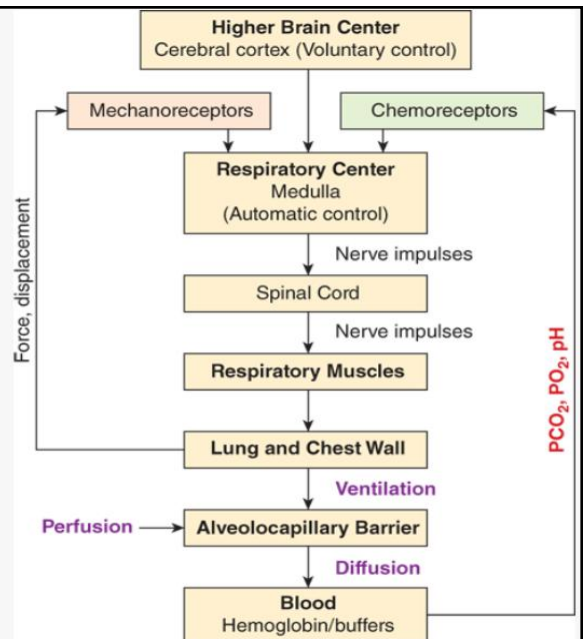
Control of breathing

1- Voluntary control:

a) **Cerebral cortex:** Descending cortical input allows voluntary control over respiration in activities like breathing pauses and depth of breathing during singing, etc.

b) **Hypothalamus:** send input signals to the respiratory center to modify the respiratory rate or depth of breathing.

Hypothalamic control is usually activated in pain, temperature or strong emotions. Fever is usually associated with increased respiratory rate and vice versa.



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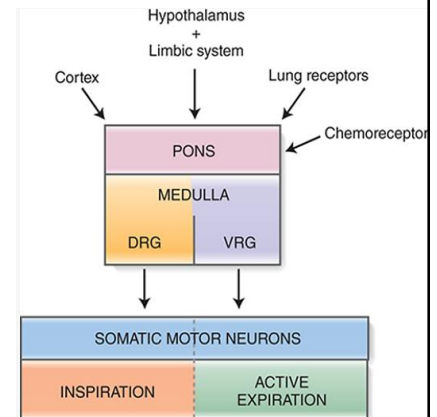
2- Automatic control:

Spontaneous ventilation is maintained by pacemaker cells in the medulla found in the respiratory center (brainstem).

Pons: control rate of involuntary respiration. Responds to feedback from multiple centers/receptors.

The medulla: control and modulate the rate of respiration through sending motor signals.

- i) Ventral respiratory group stimulate expiration. Not active in normal breathing (expiration is passive). Becomes activated when expiration becomes an active process (during exercise).
- ii) Dorsal respiratory group control inspiration and set the rhythm for breathing. Sends output to the diaphragm.



Receptors

Chemoreceptors and mechanoreceptors sense gas levels and physical changes, respectively and adjust the rate of ventilation to match the body needs.

Mechanoreceptors

Located in the lungs, joints, muscles. Activated by muscle and joint movements. Send sensory information to enable responding to different body needs, increased respiration rate during exercise.

Pulmonary Irritant receptors:

In the respiratory epithelium. Detect various irritants (like pollen, dust), send afferent sensory signals and trigger a coughing response.

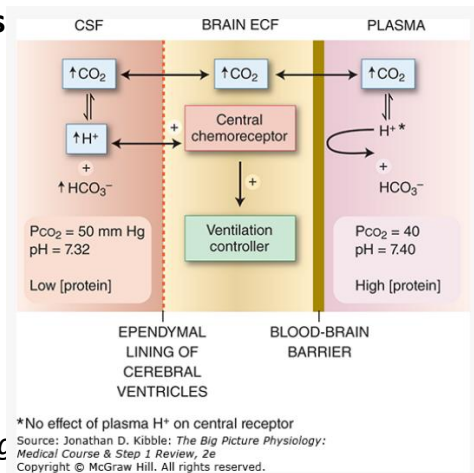
Chemoreceptors sense changes in carbon dioxide, pH or arterial oxygen and send signals to the respiratory center to adjust ventilation rate. **Carbon dioxide level is the primary driver for respiration.**

A) Central Chemoreceptors

Located in medulla, exposed to CSF. Sensitive to CSF pH changes (changes in carbon dioxide level).

Very responsive system. Critical for precise control of respiration rate.

For example, a decrease of carbon dioxide level (increase in pH) will reduce the activity of central chemoreceptors leading to a reduced respiratory drive and reduced alveolar ventilation.

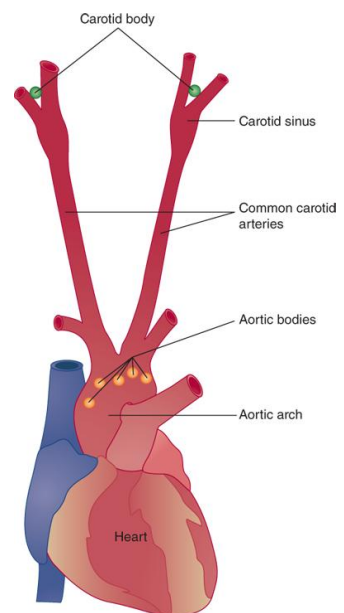


B) Peripheral Chemoreceptors

Located in the carotid and aortic bodies. Exposed to arterial blood. In response to **decreased arterial oxygen** or increased carbon dioxide levels or change in pH levels, peripheral chemoreceptors send sensory information to the dorsal respiratory group leading to increased respiration rate

Carotid and aortic chemoreceptors are more sensitive to changes in oxygen level.

For example, response to hypoxia depend on peripheral chemoreceptors.



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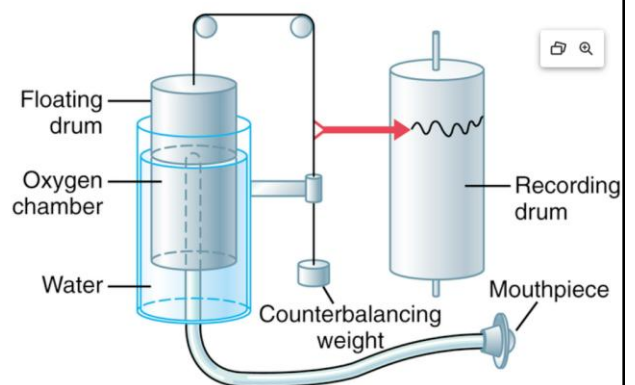
Lung Volumes and Capacity

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Spirometry is a procedure that measures the volume of air that moves in and out of the lungs during the breathing cycle.

Assess the lung volume, capacity, rates of flow, and gas exchange. Can be used to diagnose pulmonary diseases like asthma, COPD or pre-op. anesthesia risk assessment.



- Air flow rate is a quantitative measures of inspiratory and expiratory flow. At rest, the normal respiratory rate for adults is around 12 breaths/minute (12-18 breaths in adults). Higher in newborn (30 b/min) and children (20-24 b/min).
- Gas exchange determines the efficiency of CO₂-O₂ exchange, and it is measured through several methods including arterial blood gas sampling.
- **Lung volume** refer to the air volume at different phases of the respiratory cycle. Subdivided into:
 - Tidal volume
 - Inspiratory reserve volume
 - Expiratory reserve volume
 - Residual volume.

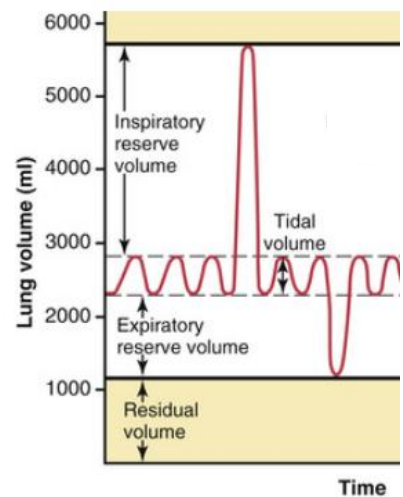
Lung volume

1- **Tidal volume**: volume of air that moves in & out of the lung during normal quiet breathing. Around 500 ml in adults.

2- **Inspiratory reserve volume**: the additional max. volume with maximal inspiratory effort (around 2 L).

3- **Expiratory reserve volume**: the additional max. volume can be exhaled below the tidal volume (around 1 L).

4- **Residual volume**: air volume that always remain in the lungs and cannot be exhaled (normally around 1-1.3 L). Can increase in obstructive lung diseases emphysema and asthma.



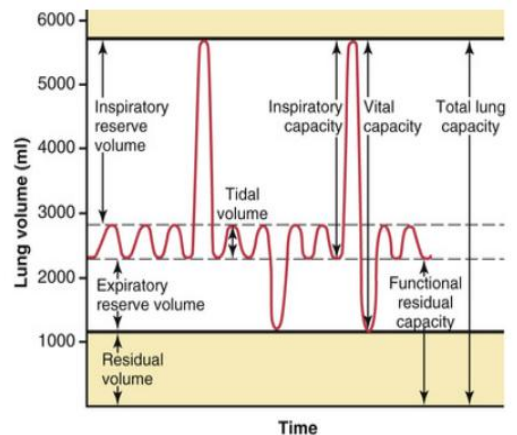
Lung Capacity

Total lung capacity: the volume of air in the lungs upon the maximum effort of inspiration (i.e, sum of all volumes). Among healthy adults, the total lung capacity is about 6 liters.

Vital lung capacity: total lung capacity without the residual volume and it refers to the maximum inspiration and maximum expiration (around 4.7 L).

The inspiratory capacity: volume of air that enters the lungs during a maximal forced inspiration beginning at the end of a normal expiration.

Functional residual capacity volume remaining in the lungs at the end of a normal tidal expiration.



Anatomic and Physiologic Dead Space

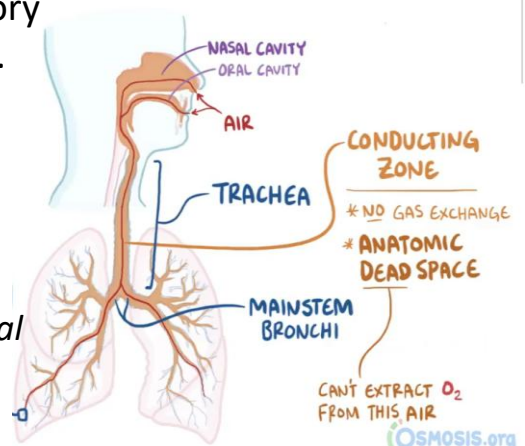
Dead space: air volume within the respiratory system that does not undergo gas exchange.

Conducting zone = “anatomic dead space” (around 150 ml of the tidal volume).

Alveolar ventilation:

Alveolar ventilation = (tidal volume – anatomical dead space) x breathing frequency.

For example, $(500 - 150) \times 12 = 4200$ ml/min



Physiologic or total dead space: total air volume that does not participate in gas exchange.

Physiologic dead space = Alveolar dead space + anatomic dead space

- In healthy adults, the “alveolar dead space” is insignificant (almost zero).
- Calculating the physiological dead space helps in determining the lung status and has diagnostic and prognostic value.
- In diseases affecting alveolar ventilation, the physiologic dead space will be greater than the anatomic dead space.

Total (minute) ventilation is the volume of air entering the lung per minute. **Needed to establish the flow rate for nitrous oxide.**

Total ventilation = tidal volume x breathing frequency

For example, 500 ml/breath x 12 breaths/min = 6000 ml/min

All inhalation anesthetics are respiratory depressants with varying degrees. They decrease the tidal volume and tend to increase the respiration rate.

Changes with Aging

With advancing age, several changes take place in the respiratory system and the lung.

Aging causes:

- 1- Loss of alveolar elastic recoil leading to a decrease in lung compliance risk factor in senile emphysema.
- 2- Alterations in chest wall structure causing decreased respiratory muscle strength.
- 3- Loss of alveolar surface area and pulmonary capillary blood volume.
- 4- Exposure to different pollutants may increase the risk of lung diseases.

Changes with Exercise

- Venous P_{CO_2} increases during exercise because excess CO_2 produced by the exercising muscle.
- Increase in ventilatory rate to match the increase in O_2 consumption and CO_2 production.
- Pulmonary blood flow increases because of increased cardiac output = more pulmonary capillaries are perfused, and more gas exchange.

Changes with Supine Position & Sedation

Supine position: increases the work of breathing through reduce lung volume (functional residual capacity), increase airway resistance, decrease alveoli compliance, and alter diaphragm mechanics. Compensated in healthy individuals but can cause orthopnea in patients with underlying cardiopulmonary disease.

Sedation: Depress CNS respiratory drive (by different levels according to sedative). Also, decrease muscle tone of upper airway tissues (risk of airway obstruction) & decrease protective airway reflexes (like coughing).

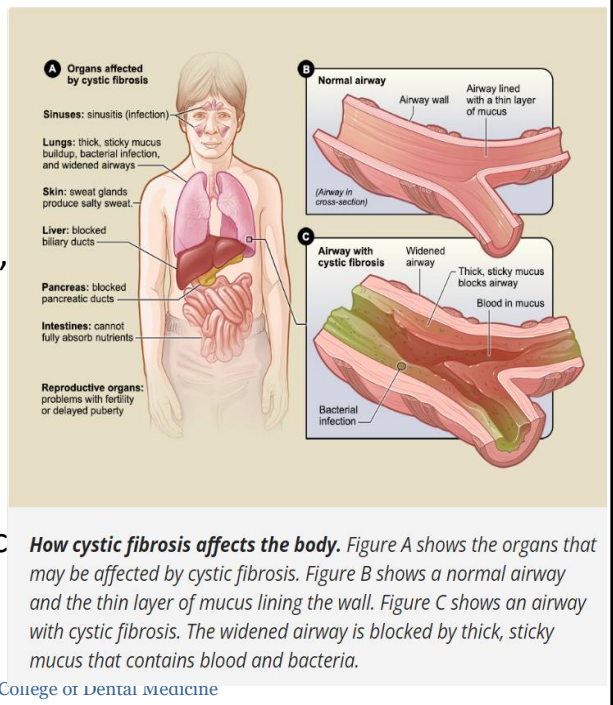
Respiratory system disorders

- Developmental disorders
- Ventilation Disorders
- Vascular disorders
- Pulmonary disorders

Developmental disorders:

Cystic Fibrosis

- Autosomal recessive disorder affects exocrine gland function in multiple systems including lungs, digestive system, sweat glands, and reproductive tract.
- Symptom severity vary among affected individuals and depend on the affected organ.
- Progressive lung disease and it is the most common cause of mortality in cystic fibrosis.



[Cystic Fibrosis - What Is Cystic Fibrosis? | NHLBI, NIH](#)

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	Cystic Fibrosis
Prevalence	Prevalent among populations of Northern European descent at a rate of 1:3500. Predicted life-expectancy is 47 years
Etiology	mutations in <i>cystic fibrosis transmembrane conductance regulator (CFTR)</i> gene on chromosome 7, responsible for transporting chloride ions.

	Cystic Fibrosis
Clinical presentation	Lungs: manifested by wheezing, chronic coughing (sometimes with bloody secretion). Pancreas: blocked pancreatic ducts causing accumulation of digestive enzymes in the pancreas, abdominal pain, diarrhea or constipation. Sweat glands: may lead to abnormal transport of chloride and sodium leading to salty sweat and skin.
Management	No definitive cure for cystic fibrosis. To prevent complications, management strategies may include antibiotics to treat and prevent lung infections anti-inflammatory medications to decrease the airway swelling.

Ventilation Disorders

Hyperventilation

Reflects the increase in the respiration rate (increased alveolar ventilation) beyond metabolic needs. Leading to hypocapnia (reduced PaCO_2) and can lead to compensated respiratory alkalosis.

Causes: Multiple factors including stress, fear, psychological factors, medications CNS stimulants, lung disease and pregnancy.

Symptoms: Including dyspnea, dizziness, chest pain, perioral numbness and paresthesia & psychological symptoms (palpitations, panic and anxiety).

Management: Stop the stressor event, sit upright, calm the affected person and advise slower breathing .

Hypoventilation

Reflects a decrease in the respiration rate (decreased alveolar ventilation) beyond metabolic needs. Lead to hypercapnia & acute respiratory acidosis.

Causes: Multiple factors including brain injury, parenchymal lung disease, severe hypoxia, depression of respiratory centers by opiates, anesthetics, airway obstruction like COPD or neuromuscular disorders or obesity-hypoventilation syndrome (BMI ≥ 30 kg/m² and PaCO₂ >45 mmHg).

Symptoms: Non-specific and depend on the underlying cause. For example, COPD patients are present with dyspnea

Management: Manage underlying cause. Supplemental oxygen, if hypoxemia is observed.

Hypoxia

Hypoxemic
hypoxia

Circulatory
hypoxia

Anemic
hypoxia

Histotoxic
hypoxia

Normal arterial blood gas level for oxygen PO_2 is 75-100 mmHg
Hypoxia <75 mmHg

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Hypoxia

Hypoxemic (hypoxic) Hypoxia: low oxygen content in the blood (tissue) due to the inability of the lungs to oxygenate blood. $\text{PaO}_2 < 60$ mm Hg stimulate the peripheral chemoreceptors to initiate a reflex increase in ventilation. Hypoxias are severe when normal respiratory functions are significantly compromised. For example, pneumonia can reduce the alveolar ventilation (hypoxemia) or hypoventilation

Stagnant hypoxia (ischemic, circulatory/hypoperfusion): inadequate blood flow to a tissue despite a normal PaO_2 and hemoglobin concentration. Secondary to circulatory defects like heart failure which can lead to inadequate oxygen delivery.

Anemic Hypoxia: decrease in oxygen-carrying capacity of RBCs (decrease of functional Hb) leading to inadequate oxygen delivery. For example, anemia, carbon monoxide poisoning and methemoglobinemia.

Histotoxic Hypoxia: normal blood oxygenation level but cells can not utilize oxygen. Can be attributed to defective cellular machinery which prevent oxygen utilization. Examples include cyanide poisoning.

- Cyanide binds to cytochrome oxidase in the respiratory chain and effectively blocks oxidative phosphorylation responsible for oxygen utilization. Hydroxocobalamin is the antidote for acute cyanide poisoning.

Carbon Dioxide

Normal PCO₂ range: 35-45 mmHg

Hypercapnia

Retention of CO₂ in the body. Characterized by significantly elevated PCO₂ and respiratory acidosis. It may initially stimulate respiration. Results from loss of ventilatory or cardiovascular functions. Symptoms of severe hypercapnia include confusion, and, eventually, coma with respiratory depression and death due to depression of the central nervous system.

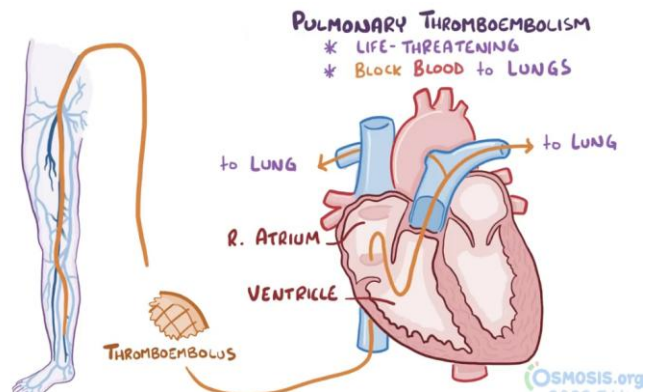
Hypocapnia

Often results from hyperventilation (acute or chronic). Can be associated with respiratory alkalosis. Severe condition can be associated with reduced cerebral blood flow leading to dizziness, and paresthesia.

Venous Thromboembolism

Venous thromboembolism (VTE) includes both deep-vein thrombosis (DVT) and pulmonary embolism (PE).

- DVT results from blood clot within large veins, usually in the legs.
- PE is the most serious complications of DVTs. A clot that have broken off and traveled to the pulmonary arterial circulation through the heart.



DVT: most common in the lower limbs. Can be asymptomatic or may present with:

- chronic leg swelling, pain and discomfort due to damage to the venous valves of the affected leg.
- Redness of the skin.

Risk factors:

- Injury to the veins (fractures, surgery)
- Slow blood flow (laying down or sitting for prolonged time).
- Hypercoagulation
- Increase estrogen
- Heart disease, obesity, age

MOST COMMON in
LOWER LIMBS
USUALLY BELOW KNEE



INFLAMED
PAIN
SWELLING
REDNESS
WARM

PATHOGENESIS & RISK FACTORS

VIRCHOW'S TRIAD

STASIS of BLOOD FLOW



HYPERCOAGULABILITY

ENDOTHELIAL INJURY

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Pulmonary embolism

Epidemiology

Incidence is 2.81/100,000

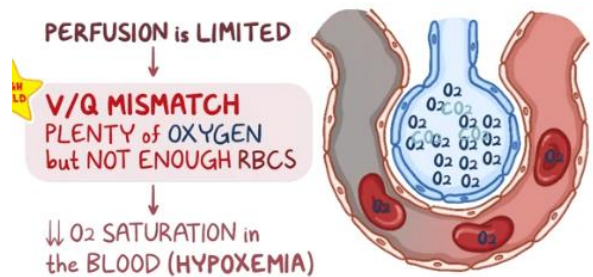
Overall incidence is higher in males.

Incidence rises with increasing age (specially in females).

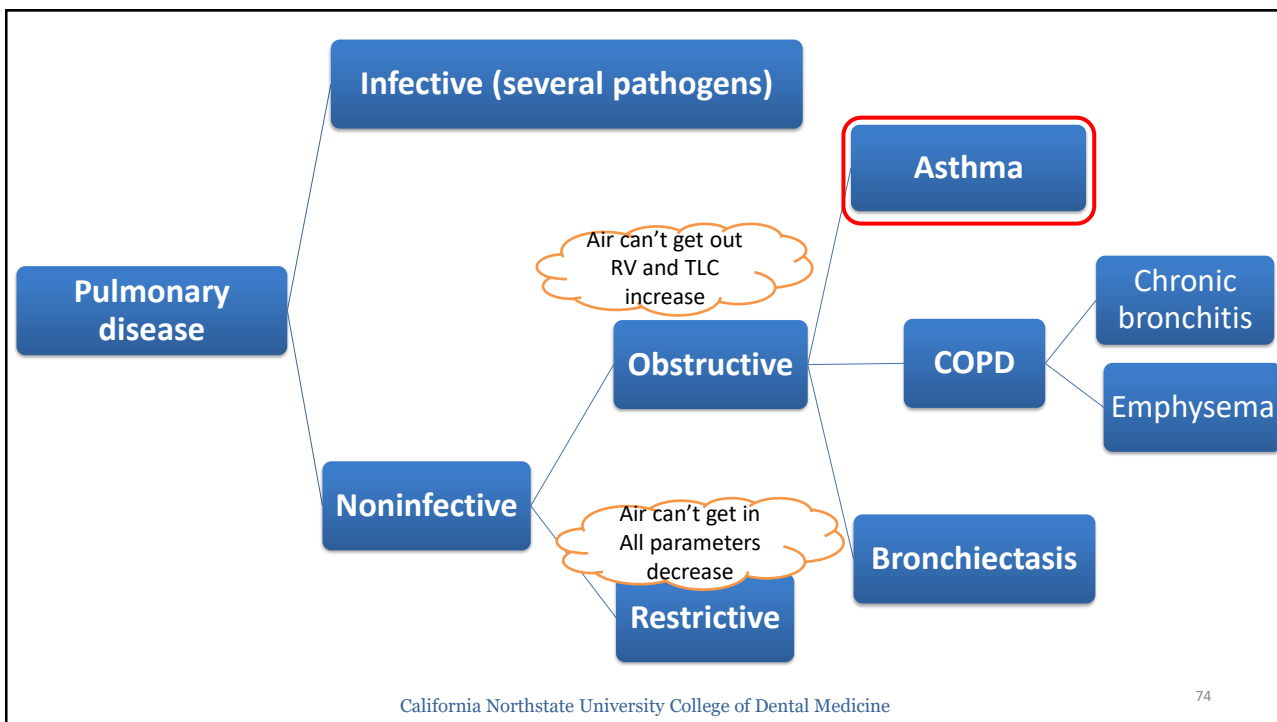
Risk factors

Include DVT (main) as well as use of pacemakers and internal cardiac defibrillator. Often diagnosed by imaging like CT pulmonary artery angiography and elevated plasma D-dimer levels.

Symptoms vary according to the occlusion. PE creates a ventilation perfusion mismatch that affects the whole body. Massive PE can lead to dyspnea, syncope, hypotension, and cyanosis and may lead to fatality. Around 45 % of acute PE patients will have acute right ventricular failure.



Management includes thrombolytic medications to dissolve the clot followed by anticoagulant treatment with warfarin (coumadin) to prevent recurrence



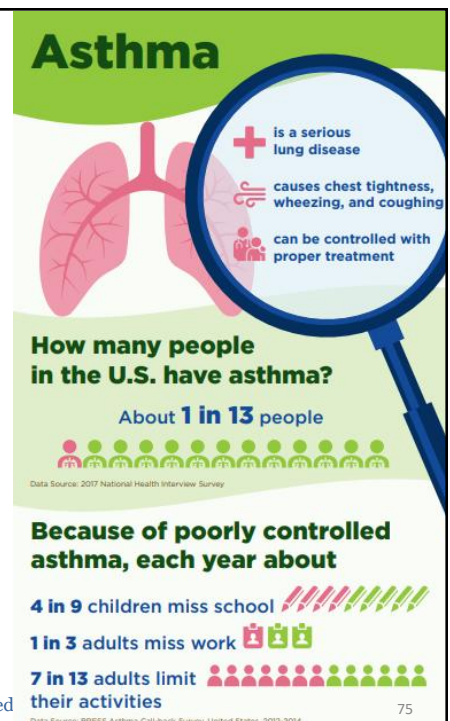
Asthma

Chronic inflammatory disorder caused by excessive airway hyperresponsiveness to stimuli. Chronic inflammation leads to structural changes, including basement membrane collagen deposition, fibrosis, airway smooth muscle hypertrophy, angiogenesis, and hyperplasia of mucus-secreting cells.

The global Burden of Disease report estimates that asthma accounts for approximately 420,000 deaths per year worldwide. It affects 1 in each 13 individuals.

[Asthma - What Is Asthma? | NHLBI, NIH](#)

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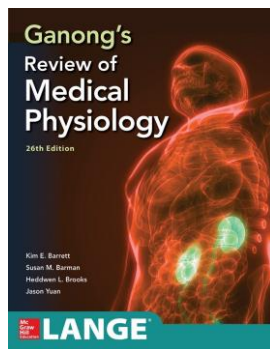
	Asthma (Intrinsic & Extrinsic)
Clinical presentation	Dry cough, chest tightness, sudden-onset dyspnea, bronchospasm and airway obstruction with breathing difficulties (wheezing). Oral?
Risk factors	Age and gender (children have higher risk and boys >girls), obesity, active and passive smoking, occupational exposures to irritants (chemical spills, fire,), exposure to allergens, positive family history for asthma. African American or Hispanic ethnicity. Atopy defines genetic predisposition to develop specific IgE antibodies against common environmental agents.
Management	Education, control trigger and several pharmacologic strategies.

[Asthma-dental](#)

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Recommended resources



OSMOSIS
from ELSEVIER



Organ Systems

Respiratory system ▾